BJT Collector-base Coupled Non-Inverting Schmitt Trigger

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1. Introduction

Schmitt triggers are used in a variety of applications from trigger inputs for microcontrollers to crude 1-bit DACs to oscillators. They can be built from a variety of components; namely, MOSFETs, BJTs, and discrete Operational Amplifiers. They work because of hysteresis, the output not being directly related to the input but rather dependent on the state of the system perviously, which allows the Schmitt trigger to output a high or low voltage based on two different threshold voltages.

2. Circuit Description

The specific Schmitt trigger circuit that we will be using is called the BJT Collector-base Coupled Non-Inverting Schmitt Trigger. It is comprised of 5 resistors and 2 BJTs in a formation that resembles a SRAM cell. The output voltage will be the voltage at the collector of Q2. The circuit diagram below shows how this circuit is built. Note that for simplicity, we will assume that R1 is equal to R2 and R3 is equal to R4.

3. Circuit Analysis

Before building this circuit, we used LTSpice to simulate the circuit which allowed us to easily change the resistance values of in the circuit to see how the circuit would respond to a sine wave input. We observed the following:

4. Experimental Setup

4.1. Experiment 1: BJT Collector-base Coupled Non-Inverting Schmitt Trigger Characterization

In this experiment, we set up the circuit as shown in Figure 1 and measured the output voltage of the circuit. We swept the voltage from V_{ee} (GND) to V_{cc} and then back to V_{ee}. We are characterizing the Upper and Lower Threshold Voltages of the circuit (represented by V_{UT} and V_{LT} respectively) and then calculating the Hysteresis Voltage (represented by V_H) by subtracting the Upper Threshold Voltage from the Lower Threshold Voltage.

$$V_H = V_{UT} - V_{LT}$$

With the same data, we also extracted the Output High and Low Voltage (represented by V_{OH} and V_{OL} respectively) by looking at what the average high voltage was and what the average low voltage was.

4.2. Experiment 2: Schmitt Trigger Frequency Response Characterization

4.3. Experiment 3: Schmitt Trigger Frequency Response Trends

5. Results

6. Analysis

7. Discussion

8. Summary

9. Sources

- Schmitt Trigger Explained (Design of Inverting and Non-inverting Schmitt Trigger using Op-Amp) Video - https://youtu.be/5-ohKRWeod4?si=8VM2gfX6_NjWiwgg

10. notes

- Has to be a gap between R1 and R2 values and R3 and R4 values; R1/R2 must be smaller than R3/R4. - Output High is determined by voltage divider ratio betwene R2 and R4 where they are a volt div - When one turns one, it sucks current from the other base of other BJT and turns the other off at the same time

Figure 1. BJT Collector-base Coupled Non-Inverting Schmitt Trigger Circuit.